BLOOR HOMES

R1/R2 DRAINAGE STATEMENT

SOUTH MARSTON, SWINDON

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INTRODUCTION

This Drainage Statement has been prepared to support the Reserved Matters Application for residential parcels R1 and R2 situated within the South Marston Eastern Villages Development.

Reference should be made to the Flood Risk Assessment submitted as part of the outline application (Ref. OUT/13/1555) prepared by Brookbanks in 2017 and the subsequent drainage strategy submitted to clear Condition 42 (Ref. S/COND/25/0950).

Parcels R1 and R2 consist of a maximum of 122 dwellings with surface water and foul discharge points provided within the Secondary Loop Road (Planning Ref. S/RES/25/0836). Attenuation for the surface water run-off is to be provided within the strategic basins and swales provided as part of the primary infrastructure works currently under construction.

SURFACE WATER

Within the drainage strategy for the site there is a total area allowance attributed to parcels R1 and R2 as follows:

R1 – 1.67 Hectares – Including Urban Creep and 15% reduction in impermeable area.
R2 – 0.50 Hectares – Including Urban Creep and 15% reduction in Impermeable area.

It is understood that the secondary loop road through the parcels forms part of the proposed R1-R2 catchment within the calculations. The calculations sheet taken directly from the Outline Condition 42 submission is included at the end of this statement.

Total allowance = 2.17 Hectares

Total impermeable area (As shown on Bloor drawing SW193-EN-502A, Surface Water catchment plan) is 2.48 Hectares including Urban Creep, consisting of the following:

R1 (Non Spine Road) Catchment 1 – 0.97 Hectares
R1 (Non Spine Road) Catchment 2 – 0.45 Hectares
Spine Road Catchment (R1 partial and R2) – 0.72 Hectares
Secondary Loop Road – 0.38 Hectares

In order to reduce the impermeable area (15% betterment) to align with the overall strategy it is suggested that 0.72 cubic metres of storage is provided for every 1 square metre of impermeable area (excluding adoptable roads). Based on the private impermeable areas (1.66 Hectares) this equates to a total permeable paving area of 650 square metres with an assumed granular storage depth of 450mm. The proposals allow for an area of 801 square metres and therefore exceeds the 15% betterment requirement.

The surface water network within Parcel R1 has been designed to cater for the 1 in 2 year return period with no flooding within the 1 in 30 year storm and minimal flooding within the 1 in 100 year storm with 40% climate change. For calculations please refer to the end of this report. Any flooding that does occur during the extreme events will be contained within the road corridor before ultimately entering public open space areas. For overland flow routes refer to Bloor drawing SW193-EN-503A.

FOUL WATER

The number of plots contributing to the foul drainage network is also in line with expectations. The foul discharge is ultimately pumped off site via the strategic foul pumping station (Planning Ref. S/RES/24/0551) with off site improvement works currently being constructed by Thames Water.

ADOPTION AND MAINTENANCE

All adoptable manholes and pipe lengths will be offered for adoption to either Thames Water or appropriate NAV via a Section 104 agreement. Road gullies will be adopted and maintained by Swindon Borough Council via a Section 38 agreement. Permeable paving will be maintained by an elected Management Company in accordance with the below schedule. The remaining drainage features will be conveyed privately and the responsibility of the occupier.

PERMEABLE AND POROUS SURFACES

REGULAR MAINTENANCE	**FREQUENCY**
Cleaning - Brush regularly and remove sweepings from all hard surfaces.	Monthly.
OCCASIONAL TASKS	**FREQUENCY**
Brush and Vacuum surface once a year to prevent silt blockage and enhance life.	Annually.
REMEDIAL WORK	**FREQUENCY**
Monitor effectiveness of permeable pavement and when water does not infiltrate immediately advise Client of possible need for reinstatement of top layers or specialist cleaning. Recent experience suggests jet washing and suction cleaning will substantially reinstate pavement to 90% efficiency.	As required.

CONCLUSION

Based on the above and supporting drawings it has been demonstrated that the proposed drainage design is fully compliant with the agreed strategy, and all downstream features are sized to accommodate the development proposals.

APPENDIX A – SWINDON EV DRAINAGE CALCULATIONS

Document reference: Approved drainage calculations (Outline condition 42).

Swindon EV Drainage Calculation **rev.A**

Impermeability Ratios		IoH 1ha		
Residential	0.55	1 Year	3.73	l/s
Education	0.45	100 Year	14.01	l/s
Mixed Use	0.85	Agreed Discharge	4.83	l/s
Retained Development	0.50			
Primary Infrastructure	0.90			
Residential with 10% urban creep	0.583			

Detention Feature	Catchment	Use	Area (ha)	Hard Area (ha)	Hard Area with 10% UC	Increase due to UC	Ex 1:100 Year discharge rate (l/s)	Hard Area (ha) Reduced by 15%	Hard Area (ha) Reduced by 15%+UC increase	Prop 1:100 Year (+40%cc) discharge rate (l/s)
Basin A	R1	Residential	3.34	1.84	1.95	0.11	25.74	1.56	1.67	7.54
	R2	Residential	1.00	0.55	0.58	0.03	7.71	0.47	0.50	2.26
	R3	Residential	0.34	0.19	0.20	0.01	2.62	0.16	0.17	0.77
	R4	Residential	2.23	1.23	1.30	0.07	17.18	1.04	1.12	5.04
	R6	Residential	3.33	1.83	1.94	0.11	25.66	1.56	1.67	7.52
	R7	Residential	1.06	0.58	0.62	0.03	8.17	0.50	0.53	2.39
		Primary Infrastructure	1.18	1.06	1.06	0.00	14.85	1.06	1.06	5.12
Storage Swale 1A	R5	Residential	1.14	0.63	0.66	0.04	8.78	0.53	0.57	2.57
Storage Swale 1B		Primary Infrastructure	1.31	1.18	1.18	0.00	16.53	1.18	1.18	5.70
	Total		**14.93**	**9.08**	**9.49**	**0.41**	**127.24**	**8.06**	**8.47**	**38.91**
Storage Swale 2	R8	Residential	2.43	1.34	1.42	0.08	18.72	1.14	1.22	5.51
	R9	Residential	1.81	1.00	1.06	0.06	13.95	0.85	0.91	4.09
	Total		**4.24**	**2.34**	**2.47**	**0.14**	**32.67**	**1.99**	**2.12**	**9.59**
Basin B	R10	Residential	7.93	4.36	4.62	0.26	61.10	3.71	3.97	17.91
	R11	Residential	3.27	1.80	1.91	0.11	25.20	1.53	1.64	7.38
	R12	Residential	1.65	0.91	0.96	0.05	12.71	0.77	0.83	3.73
	R13	Residential	2.39	1.31	1.39	0.08	18.42	1.12	1.20	5.40
	Primary School	Education	2.25	1.01	1.01	0.00	14.19	1.01	1.01	4.88
	Retained Dev	Commercial	0.30	0.17	0.17	0.01	2.31	0.17	0.18	0.82
	Mixed Use	Mixed Use	0.42	0.36	0.36	0.00	5.00	0.36	0.36	1.74
	Total		**18.21**	**9.92**	**10.42**	**0.51**	**138.93**	**8.66**	**9.17**	**41.85**
Basin C	R14	Residential	2.62	1.44	1.53	0.09	20.19	1.22	1.31	5.92
	R15	Residential	1.28	0.70	0.75	0.04	9.86	0.60	0.64	2.89
	R16	Residential	1.40	0.77	0.82	0.05	10.79	0.65	0.70	3.16
	Mixed Use	Mixed Use	0.70	0.60	0.60	0.01	8.34	0.60	0.61	2.90
	R17(part)	Residential	0.61	0.34	0.36	0.02	4.70	0.29	0.31	1.38
	R18(part)	Residential	2.35	1.29	1.37	0.08	18.11	1.10	1.18	5.31
		Primary Infrastructure	0.71	0.64	0.64	0.00	8.95	0.64	0.64	3.09
	Total		**9.67**	**5.78**	**6.05**	**0.28**	**80.94**	**5.10**	**5.38**	**24.64**
Basin D	R17(part)	Residential	7.77	4.27	4.53	0.26	59.87	3.63	3.89	17.54
	R18(part)	Residential	3.80	2.09	2.22	0.13	29.28	1.78	1.90	8.58
		Primary Infrastructure	0.68	0.61	0.61	0.00	8.57	0.61	0.61	2.96
	Total		**12.25**	**6.98**	**7.36**	**0.38**	**97.73**	**6.02**	**6.40**	**29.08**
Basin E	R19	Residential	0.50	0.28	0.29	0.02	3.85	0.23	0.25	1.13
	R20	Residential	3.06	1.68	1.78	0.10	23.58	1.43	1.53	6.91
	R21	Residential	7.41	4.08	4.32	0.24	57.10	3.46	3.71	16.73
	Mixed Use	Mixed Use	0.50	0.43	0.43	0.00	5.95	0.43	0.43	2.08
		Primary Infrastructure	0.62	0.56	0.56	0.00	7.82	0.56	0.56	2.70
	Total		**12.09**	**7.02**	**7.38**	**0.36**	**98.30**	**6.12**	**6.48**	**29.54**
Basin F	R22	Residential	2.88	1.58	1.68	0.10	22.19	1.35	1.44	6.50
	Total		**2.88**	**1.58**	**1.68**	**0.10**	**22.19**	**1.35**	**1.44**	**6.50**
Storage Swale 3		Primary Infrastructure	0.57	0.51	0.51	0.00	7.19	0.51	0.51	2.48
	Total		**0.57**	**0.51**	**0.51**	**0.00**	**7.19**	**0.51**	**0.51**	**2.48**
Storage Swale 4		Primary Infrastructure	0.46	0.41	0.41	0.00	5.80	0.41	0.41	2.00
	Total		**0.46**	**0.41**	**0.41**	**0.00**	**5.80**	**0.41**	**0.41**	**2.00**
Storage Swale 5		Primary Infrastructure	0.36	0.32	0.32	0.00	4.54	0.32	0.32	**1.56**
	Total		**0.36**	**0.32**	**0.32**	**0.00**	**4.54**	**0.32**	**0.32**	**1.56**
	Grand Total		**75.66**	**43.94**	**46.11**	**2.17**	**615.52**	**38.54**	**40.71**	**186.16**

APPENDIX B – R1/R2 DRAINAGE CALCULATIONS

Calculations prepared in accordance with the following submitted information:

- SW192-EN-501A (Drainage Layout)
- SW192-EN-502A (Drainage Catchment Plan)
- SW192-EN-503A (Surface Water Overland Flow Routes)

Network Details

Manhole Schedule

Manhole	Catchment Area (ha)	Diameter (m)	Type	CL (m)	IL (m)	Depth To Soffit (m)	Easting (m)	Northing (m)
PS17	0.000	0.180	Type E	99.996	98.450	1.396	419080.864	187251.099
S20	0.074	1.350	Type C	99.837	98.412	1.200	419085.629	187265.967
S8	0.098	1.350	Type C	99.682	98.257	1.200	419091.038	187257.289
PS3	0.000	0.180	Type E	98.813	97.720	0.993	419139.448	187287.527
S9	0.106	1.350	Type C	98.925	97.440	1.185	419134.171	187286.025
S31	0.000	0.180	Type E	98.847	97.650	1.047	419117.080	187321.075
PS1	0.000	0.180	Type E	98.982	97.900	0.982	419111.340	187314.715
S10	0.038	1.350	Type C	98.960	97.180	1.480	419112.853	187318.274
PS5	0.000	0.180	Type E	100.118	98.800	1.168	418998.663	187309.985
S1	0.127	1.350	Type C	100.196	98.573	1.401	419006.769	187297.755
PS4	0.000	0.180	Type E	100.637	99.190	1.297	419023.033	187276.143
S30	0.069	1.350	Type C	100.493	99.068	1.200	419019.473	187278.211
S2	0.106	1.200	Type B	100.373	98.490	1.658	419014.309	187286.118
PS19	0.000	0.180	Type E	100.333	99.200	1.033	419018.580	187293.644
PS20	0.000	0.180	Type E	100.119	98.850	1.169	419028.548	187300.252
PS21	0.000	0.180	Type E	99.849	98.550	1.199	419039.680	187307.631
PS6	0.000	0.180	Type E	99.574	98.300	1.124	419049.814	187314.348
PS23	0.000	0.180	Type E	99.910	98.650	1.160	419075.999	187291.313
PS22	0.000	0.180	Type E	100.028	98.750	1.178	419067.335	187286.452
S21	0.072	1.350	Type C	99.804	98.379	1.200	419069.889	187289.759
PS7	0.000	0.180	Type E	99.785	98.600	1.085	419052.544	187303.622
S3	0.159	1.350	Type C	99.489	97.790	1.399	419054.353	187312.505
S11	0.020	1.350	Type C	98.543	97.079	1.239	419090.106	187352.546
PS8	0.000	0.180	Type E	98.683	97.400	1.133	419090.675	187341.432
PS9	0.000	0.180	Type E	98.718	97.100	1.468	419099.712	187335.249
S4	0.063	1.350	Type C	98.486	96.793	1.318	419097.733	187341.244
PS16	0.000	0.180	Type E	98.134	96.470	1.514	419116.686	187359.215
S5	0.038	1.350	Type C	97.919	96.180	1.364	419119.710	187354.653
S6	0.047	1.350	Type C	97.392	95.640	1.377	419141.273	187358.787
S7	0.000	1.350	Type C	96.896	95.150	1.371	419161.924	187360.062

Pipe Schedule

Pipe Number	US Manhole	US IL (m)	DS Manhole	DS IL (m)	Shape	Dimension (m)	Length (m)	Gradient (1:x)	Roughness (mm)	US Depth To Soffit (m)	DS Depth To Soffit (m)
1.000	PS17	98.450	S8	98.332	Circ	0.15mØ	11.909	101.2	0.600	1.396	1.200
2.000	S20	98.412	S8	98.257	Circ	0.225mØ	10.225	66.1	0.600	1.200	1.200
1.001	S8	98.257	S9	97.515	Circ	0.225mØ	51.828	69.8	0.600	1.200	1.185
3.000	PS3	97.720	S9	97.640	Circ	0.1mØ	5.486	68.6	0.600	0.993	1.185
1.002	S9	97.440	S10	97.180	Circ	0.3mØ	38.658	148.7	0.600	1.185	1.480
4.000	S31	97.650	S10	97.330	Circ	0.15mØ	5.071	15.8	0.600	1.047	1.480
5.000	PS1	97.900	S10	97.380	Circ	0.1mØ	3.867	7.4	0.600	0.982	1.480
1.003	S10	97.180	S4	96.868	Circ	0.3mØ	27.499	88.1	0.600	1.480	1.318
6.000	PS5	98.800	S1	98.645	Circ	0.15mØ	14.672	94.7	0.600	1.168	1.401
6.001	S1	98.573	S2	98.490	Circ	0.225mØ	13.865	167.0	0.600	1.398	1.658
7.000	PS4	99.190	S30	99.143	Circ	0.15mØ	4.117	87.6	0.600	1.297	1.200
7.001	S30	99.068	S2	98.490	Circ	0.225mØ	9.444	16.3	0.600	1.200	1.658
6.002	S2	98.490	S3	97.865	Circ	0.225mØ	47.956	76.7	0.600	1.658	1.399
8.000	PS19	99.200	PS20	98.850	Circ	0.1mØ	11.959	34.2	0.600	1.033	1.169
8.001	PS20	98.850	PS21	98.550	Circ	0.1mØ	13.356	44.5	0.600	1.169	1.199
8.002	PS21	98.550	PS6	98.350	Circ	0.1mØ	12.158	60.8	0.600	1.199	1.124
8.003	PS6	98.300	S3	97.940	Circ	0.15mØ	4.898	13.6	0.600	1.124	1.399
9.000	PS23	98.650	S21	98.504	Circ	0.1mØ	6.305	43.3	0.600	1.160	1.200
10.000	PS22	98.750	S21	98.504	Circ	0.1mØ	4.178	17.0	0.600	1.178	1.200
9.001	S21	98.379	S3	97.865	Circ	0.225mØ	27.546	53.6	0.600	1.200	1.399
11.000	PS7	98.600	S3	97.990	Circ	0.1mØ	9.065	14.9	0.600	1.085	1.399
6.003	S3	97.790	S4	96.868	Circ	0.3mØ	52.036	56.4	0.600	1.399	1.318
12.000	S11	97.079	S4	96.943	Circ	0.225mØ	13.635	100.0	0.600	1.239	1.318
13.000	PS8	97.400	S4	97.018	Circ	0.15mØ	7.060	18.5	0.600	1.133	1.318
14.000	PS9	97.100	S4	97.018	Circ	0.15mØ	6.313	77.0	0.600	1.468	1.318
1.004	S4	96.793	S5	96.180	Circ	0.375mØ	25.745	42.0	0.600	1.318	1.364
15.000	PS16	96.470	S5	96.405	Circ	0.15mØ	5.473	84.2	0.600	1.514	1.364

Pipe Schedule

Pipe Number	US Manhole	US IL (m)	DS Manhole	DS IL (m)	Shape	Dimension (m)	Length (m)	Gradient (1:x)	Roughness (mm)	US Depth To Soffit (m)	DS Depth To Soffit (m)
1.005	S5	96.180	S6	95.640	Circ	0.375mØ	21.956	40.7	0.600	1.364	1.377
1.006	S6	95.640	S7	95.150	Circ	0.375mØ	20.690	42.2	0.600	1.377	1.371

Permeable Paving Schedule

Permeable Paving	Assigned Manhole	Effective Storage Volume (m3)	CL (m)	IL (m)	Storage Infil Rate (m/hr)	Safety Factor	Easting (m)	Northing (m)
Permeable Paving5	S1	18.015	100.196	99.816	0.00000000	2.00	419000.752	187311.095
Permeable Paving6	S30	18.408	100.493	100.113	0.00000000	2.00	419024.232	187269.711

Outfall Details

Outfall Manhole S7 : Free Discharge

Flow Control Details

Simulation Settings

FSR: M5-60=20.00, R=0.40, Locale=England and Wales

Summer (Cv: 1.00), Winter (Cv: 1.00)

Global Time of Entry: 5.0 mins

Durations (mins): 15, 30, 60, 120, 180, 240, 360, 480, 600, 720, 960, 1440, 2160, 2880, 4320, 5760, 7200, 8640, 10080

Return Periods (yrs) + Climate Change: (2, +0%), (30, +0%), (100, +40%)

Simulated Rainfall Events

Storm	Average Intensity (mm/hr)	Runoff Continuity %	Flow Continuity %	Storm	Average Intensity (mm/hr)	Runoff Continuity %	Flow Continuity %
2Yr 15Min Winter	42.301	0.00	0.00	30Yr 720Min Winter	4.687	0.00	0.00
2Yr 15Min Summer	42.301	0.00	0.00	30Yr 960Min Summer	3.743	0.00	0.00
2Yr 30Min Winter	27.017	0.00	0.00	30Yr 960Min Winter	3.743	0.00	0.00
2Yr 30Min Summer	27.017	0.00	0.00	30Yr 1440Min Summer	2.723	0.00	0.00
2Yr 60Min Winter	16.649	0.00	0.00	30Yr 1440Min Winter	2.723	0.00	0.00
2Yr 60Min Summer	16.649	0.00	0.02	30Yr 2160Min Summer	1.979	0.00	0.00
2Yr 120Min Winter	10.070	0.00	0.00	30Yr 2160Min Winter	1.979	0.00	0.00
2Yr 120Min Summer	10.070	0.00	0.00	30Yr 2880Min Summer	1.577	0.00	0.00
2Yr 180Min Summer	7.475	0.00	0.00	30Yr 2880Min Winter	1.577	0.00	0.00
2Yr 180Min Winter	7.475	0.00	0.00	30Yr 4320Min Summer	1.143	0.00	0.00
2Yr 240Min Summer	6.048	0.00	0.00	30Yr 4320Min Winter	1.143	0.00	0.00
2Yr 240Min Winter	6.048	0.00	0.00	30Yr 5760Min Summer	0.910	0.00	0.00
2Yr 360Min Summer	4.479	0.00	0.00	30Yr 5760Min Winter	0.910	0.00	0.00
2Yr 360Min Winter	4.479	0.00	0.00	30Yr 7200Min Summer	0.762	0.00	0.00
2Yr 480Min Summer	3.616	0.00	0.00	30Yr 7200Min Winter	0.762	0.00	0.00
2Yr 480Min Winter	3.616	0.00	0.00	30Yr 8640Min Summer	0.659	0.00	0.00
2Yr 600Min Summer	3.062	0.00	0.00	30Yr 8640Min Winter	0.659	0.00	0.00
2Yr 600Min Winter	3.062	0.00	0.00	30Yr 10080Min Summer	0.583	0.00	0.00
2Yr 720Min Summer	2.673	0.00	0.00	30Yr 10080Min Winter	0.583	0.00	0.00
2Yr 720Min Winter	2.673	0.00	0.00	100Yr+40% 15Min Summer	138.153	0.00	0.06
2Yr 960Min Summer	2.156	0.00	0.00	100Yr+40% 15Min Winter	138.153	0.00	0.07
2Yr 960Min Winter	2.156	0.00	0.00	100Yr+40% 30Min Summer	90.705	0.00	-0.03
2Yr 1440Min Winter	1.593	0.00	0.00	100Yr+40% 30Min Winter	90.705	0.00	0.01
2Yr 1440Min Summer	1.593	0.00	0.00	100Yr+40% 60Min Summer	56.713	0.00	-0.01
2Yr 2160Min Winter	1.177	0.00	0.00	100Yr+40% 60Min Winter	56.713	0.00	-0.07
2Yr 2160Min Summer	1.177	0.00	0.00	100Yr+40% 120Min Summer	34.246	0.00	-0.04
2Yr 2880Min Summer	0.949	0.00	0.00	100Yr+40% 120Min Winter	34.246	0.00	0.01
2Yr 2880Min Winter	0.949	0.00	0.00	100Yr+40% 180Min Summer	25.149	0.00	0.01
2Yr 4320Min Summer	0.701	0.00	0.00	100Yr+40% 180Min Winter	25.149	0.00	0.01
2Yr 4320Min Winter	0.701	0.00	0.00	100Yr+40% 240Min Summer	20.078	0.00	0.01
2Yr 5760Min Summer	0.566	0.00	0.00	100Yr+40% 240Min Winter	20.078	0.00	0.01
2Yr 5760Min Winter	0.566	0.00	0.00	100Yr+40% 360Min Summer	14.585	0.00	0.00
2Yr 7200Min Summer	0.479	0.00	0.00	100Yr+40% 360Min Winter	14.585	0.00	0.00
2Yr 7200Min Winter	0.479	0.00	0.00	100Yr+40% 480Min Summer	11.622	0.00	0.00
2Yr 8640Min Summer	0.419	0.00	0.00	100Yr+40% 480Min Winter	11.622	0.00	0.00
2Yr 8640Min Winter	0.419	0.00	0.00	100Yr+40% 600Min Summer	9.738	0.00	0.00
2Yr 10080Min Winter	0.373	0.00	0.00	100Yr+40% 600Min Winter	9.738	0.00	0.00
2Yr 10080Min Summer	0.373	0.00	0.00	100Yr+40% 720Min Summer	8.424	0.00	0.00
30Yr 15Min Summer	76.035	0.00	-0.18	100Yr+40% 720Min Winter	8.424	0.00	0.00
30Yr 15Min Winter	76.035	0.00	-0.24	100Yr+40% 960Min Summer	6.697	0.00	0.00
30Yr 30Min Summer	49.499	0.00	-0.12	100Yr+40% 960Min Winter	6.697	0.00	0.00
30Yr 30Min Winter	49.499	0.00	0.01	100Yr+40% 1440Min Summer	4.839	0.00	0.00
30Yr 60Min Summer	30.811	0.00	0.01	100Yr+40% 1440Min Winter	4.839	0.00	0.00
30Yr 60Min Winter	30.811	0.00	0.00	100Yr+40% 2160Min Summer	3.490	0.00	0.00
30Yr 120Min Summer	18.615	0.00	0.01	100Yr+40% 2160Min Winter	3.490	0.00	0.00
30Yr 120Min Winter	18.615	0.00	0.00	100Yr+40% 2880Min Summer	2.766	0.00	0.00
30Yr 180Min Summer	13.715	0.00	0.01	100Yr+40% 2880Min Winter	2.766	0.00	0.00
30Yr 180Min Winter	13.715	0.00	0.00	100Yr+40% 4320Min Summer	1.989	0.00	0.00
30Yr 240Min Summer	10.995	0.00	0.01	100Yr+40% 4320Min Winter	1.989	0.00	0.00
30Yr 240Min Winter	10.995	0.00	0.00	100Yr+40% 5760Min Summer	1.573	0.00	0.00
30Yr 360Min Summer	8.034	0.00	0.00	100Yr+40% 5760Min Winter	1.573	0.00	0.00
30Yr 360Min Winter	8.034	0.00	0.00	100Yr+40% 7200Min Summer	1.311	0.00	0.00
30Yr 480Min Summer	6.428	0.00	0.00	100Yr+40% 7200Min Winter	1.311	0.00	0.00
30Yr 480Min Winter	6.428	0.00	0.00	100Yr+40% 8640Min Summer	1.129	0.00	0.00
30Yr 600Min Summer	5.404	0.00	0.00	100Yr+40% 8640Min Winter	1.129	0.00	0.00

Simulated Rainfall Events

Storm	Average Intensity (mm/hr)	Runoff Continuity %	Flow Continuity %	Storm	Average Intensity (mm/hr)	Runoff Continuity %	Flow Continuity %
30Yr 600Min Winter	5.404	0.00	0.00	100Yr+40% 10080Min Winter	0.994	0.00	0.00
30Yr 720Min Summer	4.687	0.00	0.00	100Yr+40% 10080Min Summer	0.994	0.00	0.00

Simulation Results

Return Period Yrs: 2.0

Climate Change %: 0

Manholes

Manhole	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
PS17	-	0	98.450	0.000	0.000		OK
S20	15 min Summer	8	98.495	0.083	18.596		OK
S8	15 min Summer	8	98.392	0.134	43.084		OK
PS3	-	0	97.720	0.000	0.000		OK
S9	15 min Summer	9	97.630	0.190	67.777		OK
S31	-	0	97.650	0.000	0.000		OK
PS1	-	0	97.900	0.000	0.000		OK
S10	15 min Summer	9	97.355	0.175	77.392		OK
PS5	15 min Summer	10	98.832	0.032	0.000		OK
S1	15 min Summer	10	98.832	0.259	26.324		Surcharged
PS4	-	0	99.190	0.000	0.000		OK
S30	15 min Summer	8	99.123	0.055	17.206		OK
S2	15 min Summer	10	98.794	0.304	63.320		Surcharged
PS19	-	0	99.200	0.000	0.000		OK
PS20	-	0	98.850	0.000	0.000		OK
PS21	-	0	98.550	0.000	0.000		OK
PS6	-	0	98.300	0.000	0.000		OK
PS23	-	0	98.650	0.000	0.000		OK
PS22	-	0	98.750	0.000	0.000		OK
S21	15 min Summer	8	98.456	0.076	18.099		OK
PS7	-	0	98.600	0.000	0.000		OK
S3	15 min Summer	9	97.993	0.203	121.688		OK
S11	15 min Summer	8	97.126	0.047	5.066		OK
PS8	-	0	97.400	0.000	0.000		OK
PS9	-	0	97.100	0.000	0.000		OK
S4	15 min Summer	9	97.022	0.229	219.872		OK
PS16	-	0	96.470	0.000	0.000		OK
S5	15 min Summer	9	96.413	0.233	229.167		OK
S6	15 min Summer	9	95.884	0.244	240.569		OK
S7	15 min Summer	9	95.394	0.244	240.782		Outfall

Conduits

Pipe No.	Critical Storm	Peak (mins)	US Manhole	DS Manhole	Flow Depth (m)	Max Velocity (m/s)	Max Flow (l/s)	Flow / Capacity	Status
1.000	15 min Summer	8	PS17	S8	0.030	0.000	0.000	0.000	OK
2.000	15 min Summer	8	S20	S8	0.109	0.978	18.494	0.289	OK
1.001	15 min Summer	9	S8	S9	0.135	1.710	42.412	0.681	OK
3.000	15 min Winter	0	PS3	S9	0.000	0.000	0.000	0.000	OK
1.002	15 min Summer	9	S9	S10	0.182	1.516	68.204	0.751	OK
4.000	15 min Summer	9	S31	S10	0.012	0.000	0.000	0.000	OK
5.000	15 min Winter	0	PS1	S10	0.000	0.000	0.000	0.000	OK
1.003	15 min Summer	9	S10	S4	0.175	1.825	77.975	0.659	OK
6.000	15 min Summer	10	PS5	S1	0.091	0.154	0.744	0.041	OK
6.001	15 min Summer	9	S1	S2	0.225	0.881	28.584	0.714	OK
7.000	15 min Winter	0	PS4	S30	0.000	0.000	0.000	0.000	OK
7.001	15 min Summer	8	S30	S2	0.140	0.780	17.164	0.133	OK
6.002	15 min Summer	10	S2	S3	0.216	1.693	66.358	1.118	OK
8.000	15 min Winter	0	PS19	PS20	0.000	0.000	0.000	0.000	OK
8.001	15 min Winter	0	PS20	PS21	0.000	0.000	0.000	0.000	OK
8.002	15 min Winter	0	PS21	PS6	0.000	0.000	0.000	0.000	OK
8.003	15 min Summer	9	PS6	S3	0.027	0.000	0.000	0.000	OK
9.000	15 min Winter	0	PS23	S21	0.000	0.000	0.000	0.000	OK
10.000	15 min Winter	0	PS22	S21	0.000	0.000	0.000	0.000	OK
9.001	15 min Summer	8	S21	S3	0.102	1.226	17.974	0.253	OK
11.000	15 min Summer	9	PS7	S3	0.002	0.000	0.000	0.000	OK
6.003	15 min Summer	9	S3	S4	0.203	2.387	121.808	0.822	OK

Conduits

Pipe No.	Critical Storm	Peak (mins)	US Manhole	DS Manhole	Flow Depth (m)	Max Velocity (m/s)	Max Flow (l/s)	Flow / Capacity	Status
12.000	15 min Summer	9	S11	S4	0.063	0.732	5.020	0.097	OK
13.000	15 min Summer	9	PS8	S4	0.002	0.000	0.000	0.000	OK
14.000	15 min Summer	9	PS9	S4	0.002	0.000	0.000	0.000	OK
1.004	15 min Summer	9	S4	S5	0.231	3.083	219.978	0.711	OK
15.000	15 min Summer	9	PS16	S5	0.004	0.000	0.000	0.000	OK
1.005	15 min Summer	9	S5	S6	0.238	3.096	229.251	0.729	OK
1.006	15 min Summer	9	S6	S7	0.244	3.167	240.782	0.780	OK

Permeable Paving Storage

Permeable Paving	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
Permeable Paving5	60 min Summer	44	99.869	0.052	1.439		OK
Permeable Paving6	60 min Summer	44	100.165	0.052	1.434		OK

Return Period Yrs: 30.0

Climate Change %: 0

Manholes

Manhole	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
PS17	15 min Summer	10	98.634	0.184	0.000		Surcharged
S20	15 min Summer	10	98.663	0.251	26.831		Surcharged
S8	15 min Summer	10	98.635	0.378	64.418		Surcharged
PS3	15 min Summer	9	97.797	0.077	0.257		OK
S9	15 min Summer	10	97.799	0.359	105.524		Surcharged
S31	-	0	97.650	0.000	0.000		OK
PS1	-	0	97.900	0.000	0.000		OK
S10	15 min Summer	10	97.495	0.315	116.526		Surcharged
PS5	15 min Summer	10	99.834	1.034	2.016		Flood Risk
S1	15 min Summer	10	99.832	1.259	45.848		Surcharged
PS4	15 min Summer	10	99.768	0.578	2.214		Surcharged
S30	15 min Summer	10	99.769	0.701	25.088		Surcharged
S2	15 min Summer	10	99.755	1.265	97.685		Surcharged
PS19	-	0	99.200	0.000	0.000		OK
PS20	-	0	98.850	0.000	0.000		OK
PS21	-	0	98.550	0.000	0.000		OK
PS6	15 min Summer	10	98.363	0.063	0.073		OK
PS23	-	0	98.650	0.000	0.000		OK
PS22	-	0	98.750	0.000	0.000		OK
S21	15 min Summer	8	98.485	0.106	32.580		OK
PS7	-	0	98.600	0.000	0.000		OK
S3	15 min Summer	10	98.361	0.571	174.255		Surcharged
S11	15 min Summer	10	97.210	0.131	7.309		OK
PS8	-	0	97.400	0.000	0.000		OK
PS9	15 min Summer	10	97.237	0.137	0.478		OK
S4	15 min Summer	10	97.225	0.432	314.952		Surcharged
PS16	15 min Summer	9	96.656	0.186	1.104		Surcharged
S5	15 min Summer	9	96.640	0.460	319.439		Surcharged
S6	15 min Summer	9	96.104	0.464	345.122		Surcharged
S7	15 min Summer	8	95.525	0.375	339.556		Outfall

Conduits

Pipe No.	Critical Storm	Peak (mins)	US Manhole	DS Manhole	Flow Depth (m)	Max Velocity (m/s)	Max Flow (l/s)	Flow / Capacity	Status
1.000	15 min Summer	11	PS17	S8	0.150	0.213	3.069	0.174	Surcharged
2.000	15 min Summer	9	S20	S8	0.225	1.079	31.531	0.493	OK
1.001	15 min Winter	9	S8	S9	0.225	1.864	67.542	1.085	OK
3.000	15 min Summer	11	PS3	S9	0.088	0.371	1.766	0.242	OK
1.002	15 min Summer	10	S9	S10	0.300	1.608	110.939	1.221	Surcharged
4.000	15 min Summer	10	S31	S10	0.075	0.000	0.000	0.000	OK
5.000	15 min Summer	10	PS1	S10	0.050	0.000	0.000	0.000	OK
1.003	15 min Summer	10	S10	S4	0.300	1.913	124.997	1.056	Surcharged
6.000	15 min Winter	14	PS5	S1	0.150	0.345	6.091	0.334	Surcharged
6.001	30 min Summer	16	S1	S2	0.225	1.015	40.368	1.008	OK
7.000	15 min Winter	12	PS4	S30	0.150	0.350	3.510	0.185	Surcharged
7.001	15 min Winter	9	S30	S2	0.225	0.968	26.632	0.206	OK
6.002	15 min Winter	11	S2	S3	0.225	2.253	89.590	1.509	OK
8.000	15 min Summer	0	PS19	PS20	0.000	0.000	0.000	0.000	OK
8.001	15 min Summer	0	PS20	PS21	0.000	0.000	0.000	0.000	OK
8.002	15 min Summer	10	PS21	PS6	0.006	0.000	0.000	0.000	OK
8.003	15 min Summer	11	PS6	S3	0.106	0.075	0.728	0.015	OK
9.000	15 min Summer	0	PS23	S21	0.000	0.000	0.000	0.000	OK
10.000	15 min Summer	0	PS22	S21	0.000	0.000	0.000	0.000	OK
9.001	15 min Summer	8	S21	S3	0.165	1.200	32.387	0.455	OK
11.000	15 min Summer	8	PS7	S3	0.050	0.000	0.000	0.000	OK
6.003	15 min Summer	10	S3	S4	0.300	2.457	168.283	1.136	Surcharged
12.000	15 min Summer	10	S11	S4	0.178	0.750	10.926	0.210	OK
13.000	15 min Summer	9	PS8	S4	0.075	0.000	0.000	0.000	OK
14.000	15 min Summer	12	PS9	S4	0.143	0.296	1.168	0.058	OK
1.004	15 min Summer	9	S4	S5	0.375	3.185	315.841	1.021	Surcharged

Conduits

Pipe No.	Critical Storm	Peak (mins)	US Manhole	DS Manhole	Flow Depth (m)	Max Velocity (m/s)	Max Flow (l/s)	Flow / Capacity	Status
15.000	15 min Winter	10	PS16	S5	0.150	0.257	1.255	0.065	Surcharged
1.005	15 min Winter	9	S5	S6	0.375	3.165	324.868	1.033	Surcharged
1.006	15 min Winter	10	S6	S7	0.375	3.244	339.239	1.099	Surcharged

Permeable Paving Storage

Permeable Paving	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
Permeable Paving5	60 min Summer	46	99.907	0.091	4.102		OK
Permeable Paving6	60 min Summer	47	100.203	0.090	3.849		OK

Return Period Yrs: 100.0

Climate Change %: 40

Manholes

Manhole	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
PS17	15 min Winter	9	99.724	1.274	1.345		Flood Risk
S20	15 min Summer	9	99.806	1.394	58.466		Flood Risk
S8	15 min Summer	9	99.682	1.425	140.199	6.713	Flood
PS3	15 min Summer	11	98.784	1.064	0.000		Flood Risk
S9	15 min Summer	10	98.782	1.342	139.890		Flood Risk
S31	15 min Summer	11	98.346	0.696	0.000		Surcharged
PS1	15 min Summer	11	98.347	0.447	0.000		Surcharged
S10	15 min Summer	11	98.344	1.164	138.484		Surcharged
PS5	30 min Winter	17	100.118	1.318	13.139	5.322	Flood
S1	15 min Summer	8	100.196	1.623	149.032	22.957	Flood
PS4	15 min Summer	8	100.410	1.220	0.000		Flood Risk
S30	15 min Summer	8	100.408	1.340	58.820		Flood Risk
S2	15 min Summer	8	100.315	1.825	135.758		Flood Risk
PS19	15 min Summer	12	99.316	0.116	0.000		Surcharged
PS20	15 min Summer	12	99.299	0.449	0.825		Surcharged
PS21	15 min Summer	11	99.320	0.770	3.899		Surcharged
PS6	15 min Summer	10	99.357	1.057	8.982		Flood Risk
PS23	15 min Summer	11	99.515	0.865	0.000		Surcharged
PS22	15 min Summer	11	99.513	0.763	0.000		Surcharged
S21	15 min Summer	10	99.510	1.130	47.821		Flood Risk
PS7	15 min Summer	11	99.365	0.765	0.000		Surcharged
S3	15 min Summer	10	99.367	1.577	215.885		Flood Risk
S11	15 min Summer	11	97.929	0.850	8.168		Surcharged
PS8	15 min Summer	11	97.929	0.529	0.000		Surcharged
PS9	15 min Summer	11	97.928	0.828	0.000		Surcharged
S4	15 min Summer	11	97.927	1.134	364.943		Surcharged
PS16	15 min Summer	11	97.107	0.637	0.000		Surcharged
S5	15 min Summer	11	97.106	0.926	384.937		Surcharged
S6	15 min Summer	10	96.341	0.701	411.807		Surcharged
S7	15 min Summer	6	95.525	0.375	324.103		Outfall

Conduits

Pipe No.	Critical Storm	Peak (mins)	US Manhole	DS Manhole	Flow Depth (m)	Max Velocity (m/s)	Max Flow (l/s)	Flow / Capacity	Status
1.000	15 min Winter	15	PS17	S8	0.150	0.406	7.176	0.407	Surcharged
2.000	15 min Winter	9	S20	S8	0.225	1.310	52.081	0.814	OK
1.001	15 min Winter	9	S8	S9	0.225	1.993	79.261	1.273	OK
3.000	15 min Winter	16	PS3	S9	0.100	0.630	4.949	0.678	Surcharged
1.002	15 min Winter	9	S9	S10	0.300	1.824	128.949	1.419	Surcharged
4.000	15 min Winter	15	S31	S10	0.150	0.226	3.847	0.086	Surcharged
5.000	15 min Winter	14	PS1	S10	0.100	0.445	3.358	0.150	Surcharged
1.003	15 min Winter	12	S10	S4	0.300	1.970	139.229	1.177	Surcharged
6.000	15 min Winter	16	PS5	S1	0.150	0.331	5.854	0.321	Surcharged
6.001	15 min Winter	7	S1	S2	0.225	1.139	45.283	1.131	OK
7.000	15 min Winter	16	PS4	S30	0.150	0.462	5.842	0.308	Surcharged
7.001	15 min Winter	8	S30	S2	0.225	1.126	44.778	0.346	OK
6.002	60 min Winter	37	S2	S3	0.225	2.431	96.653	1.628	OK
8.000	15 min Summer	13	PS19	PS20	0.100	0.306	1.656	0.159	Surcharged
8.001	15 min Winter	14	PS20	PS21	0.100	0.395	3.104	0.342	Surcharged
8.002	15 min Winter	15	PS21	PS6	0.100	0.867	6.812	0.878	Surcharged
8.003	15 min Winter	15	PS6	S3	0.150	0.670	11.847	0.244	Surcharged
9.000	15 min Winter	14	PS23	S21	0.100	0.561	4.407	0.478	Surcharged
10.000	15 min Winter	14	PS22	S21	0.100	0.569	4.467	0.302	Surcharged
9.001	15 min Winter	8	S21	S3	0.225	1.229	40.519	0.569	OK
11.000	15 min Winter	14	PS7	S3	0.100	0.520	4.077	0.258	Surcharged
6.003	15 min Winter	10	S3	S4	0.300	2.683	189.629	1.280	Surcharged
12.000	15 min Winter	8	S11	S4	0.225	0.731	13.213	0.254	OK
13.000	15 min Winter	15	PS8	S4	0.150	0.271	3.378	0.081	Surcharged
14.000	15 min Winter	16	PS9	S4	0.150	0.576	3.931	0.194	Surcharged
1.004	15 min Winter	12	S4	S5	0.375	3.284	362.742	1.172	Surcharged

Conduits

Pipe No.	Critical Storm	Peak (mins)	US Manhole	DS Manhole	Flow Depth (m)	Max Velocity (m/s)	Max Flow (l/s)	Flow / Capacity	Status
15.000	30 min Winter	26	PS16	S5	0.150	0.936	7.536	0.390	Surcharged
1.005	15 min Winter	11	S5	S6	0.375	3.443	380.306	1.209	Surcharged
1.006	15 min Winter	11	S6	S7	0.375	3.644	402.419	1.304	Surcharged

Permeable Paving Storage

Permeable Paving	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
Permeable Paving5	60 min Summer	47	100.050	0.233	8.179		OK
Permeable Paving6	60 min Summer	47	100.285	0.172	7.711		OK

Network Details

Manhole Schedule

Manhole	Catchment Area (ha)	Diameter (m)	Type	CL (m)	IL (m)	Depth To Soffit (m)	Easting (m)	Northing (m)
PS12	0.076	0.180	Type E	98.809	97.250	1.409	419052.054	187374.357
SA12	0.123	1.350	Type C	98.530	97.105	1.200	419053.183	187381.310
PS10	0.000	0.180	Type E	98.785	97.350	1.335	419067.414	187373.575
PS11	0.000	0.180	Type E	98.675	97.350	1.225	419079.195	187380.765
SA22	0.083	1.350	Type C	98.539	97.119	1.195	419073.543	187377.636
SA13	0.023	1.350	Type C	98.299	96.797	1.202	419065.706	187389.751
SA14	0.073	1.350	Type C	97.632	96.015	1.317	419102.397	187414.052
SA15	0.020	1.350	Type C	96.906	95.330	1.276	419142.422	187440.554
SA16	0.036	1.350	Type C	96.789	95.230	1.259	419150.718	187442.483
PS14	0.000	0.180	Type E	97.142	96.250	0.792	419154.460	187431.554
PS15	0.000	0.180	Type E	96.770	95.750	0.920	419165.906	187441.435
SA17	0.062	1.350	Type C	96.598	95.040	1.258	419162.550	187437.270
SA18	0.000	1.350	Type C	96.339	94.706	1.333	419170.482	187420.916
SA19	0.000	1.350	Type C	96.217	94.548	1.369	419181.819	187408.583

Pipe Schedule

Pipe Number	US Manhole	US IL (m)	DS Manhole	DS IL (m)	Shape	Dimension (m)	Length (m)	Gradient (1:x)	Roughness (mm)	US Depth To Soffit (m)	DS Depth To Soffit (m)
1.000	PS12	97.250	SA12	97.180	Circ	0.15mØ	7.044	100.9	0.600	1.409	1.200
1.001	SA12	97.105	SA13	96.872	Circ	0.225mØ	15.101	64.7	0.600	1.200	1.202
2.000	PS10	97.350	SA22	97.244	Circ	0.1mØ	7.353	69.5	0.600	1.335	1.195
3.000	PS11	97.350	SA22	97.244	Circ	0.1mØ	6.461	61.1	0.600	1.225	1.195
2.001	SA22	97.119	SA13	96.872	Circ	0.225mØ	14.429	58.4	0.600	1.195	1.202
1.002	SA13	96.797	SA14	96.015	Circ	0.3mØ	44.009	56.3	0.600	1.202	1.317
1.003	SA14	96.015	SA15	95.330	Circ	0.3mØ	48.004	70.1	0.600	1.317	1.276
1.004	SA15	95.330	SA16	95.230	Circ	0.3mØ	8.517	85.2	0.600	1.276	1.259
1.005	SA16	95.230	SA17	95.040	Circ	0.3mØ	12.930	68.1	0.600	1.259	1.258
4.000	PS14	96.250	SA17	95.240	Circ	0.1mØ	9.906	9.8	0.600	0.792	1.258
5.000	PS15	95.750	SA17	95.240	Circ	0.1mØ	5.349	10.5	0.600	0.920	1.258
1.006	SA17	95.040	SA18	94.706	Circ	0.3mØ	18.176	54.5	0.600	1.258	1.333
1.007	SA18	94.706	SA19	94.548	Circ	0.3mØ	16.752	105.9	0.600	1.333	1.369

Permeable Paving Schedule

Permeable Paving	Assigned Manhole	Effective Storage Volume (m3)	CL (m)	IL (m)	Storage Infil Rate (m/hr)	Safety Factor	Easting (m)	Northing (m)
Permeable Paving4	PS12	38.114	98.809	98.429	0.00000000	2.00	419026.679	187363.983

Outfall Details

Outfall Manhole SA19 : Free Discharge

Flow Control Details

Simulation Settings

FSR: M5-60=20.00, R=0.40, Locale=England and Wales

Summer (Cv: 1.00), Winter (Cv: 1.00)

Global Time of Entry: 5.0 mins

Durations (mins): 15, 30, 60, 120, 180, 240, 360, 480, 600, 720, 960, 1440, 2160, 2880, 4320, 5760, 7200, 8640, 10080

Return Periods (yrs) + Climate Change: (2, +0%), (30, +0%), (100, +40%)

Simulated Rainfall Events

Storm	Average Intensity (mm/hr)	Runoff Continuity %	Flow Continuity %	Storm	Average Intensity (mm/hr)	Runoff Continuity %	Flow Continuity %
2Yr 15Min Winter	42.301	0.00	0.00	30Yr 720Min Winter	4.687	0.00	0.00
2Yr 15Min Summer	42.301	0.00	0.00	30Yr 960Min Summer	3.743	0.00	0.00
2Yr 30Min Winter	27.017	0.00	0.00	30Yr 960Min Winter	3.743	0.00	0.00
2Yr 30Min Summer	27.017	0.00	0.00	30Yr 1440Min Summer	2.723	0.00	0.00
2Yr 60Min Winter	16.649	0.00	0.00	30Yr 1440Min Winter	2.723	0.00	0.00
2Yr 60Min Summer	16.649	0.00	0.00	30Yr 2160Min Summer	1.979	0.00	0.00
2Yr 120Min Winter	10.070	0.00	0.00	30Yr 2160Min Winter	1.979	0.00	0.00
2Yr 120Min Summer	10.070	0.00	0.00	30Yr 2880Min Summer	1.577	0.00	0.00
2Yr 180Min Summer	7.475	0.00	0.00	30Yr 2880Min Winter	1.577	0.00	0.00
2Yr 180Min Winter	7.475	0.00	0.00	30Yr 4320Min Summer	1.143	0.00	0.00
2Yr 240Min Summer	6.048	0.00	0.00	30Yr 4320Min Winter	1.143	0.00	0.00
2Yr 240Min Winter	6.048	0.00	0.00	30Yr 5760Min Summer	0.910	0.00	0.00
2Yr 360Min Summer	4.479	0.00	0.00	30Yr 5760Min Winter	0.910	0.00	0.00
2Yr 360Min Winter	4.479	0.00	0.00	30Yr 7200Min Summer	0.762	0.00	0.00
2Yr 480Min Summer	3.616	0.00	0.00	30Yr 7200Min Winter	0.762	0.00	0.00
2Yr 480Min Winter	3.616	0.00	0.00	30Yr 8640Min Summer	0.659	0.00	0.00
2Yr 600Min Summer	3.062	0.00	0.00	30Yr 8640Min Winter	0.659	0.00	0.00
2Yr 600Min Winter	3.062	0.00	0.00	30Yr 10080Min Summer	0.583	0.00	0.00
2Yr 720Min Summer	2.673	0.00	0.00	30Yr 10080Min Winter	0.583	0.00	0.00
2Yr 720Min Winter	2.673	0.00	0.00	100Yr+40% 15Min Summer	138.153	0.00	-0.51
2Yr 960Min Summer	2.156	0.00	0.00	100Yr+40% 15Min Winter	138.153	0.00	-0.51
2Yr 960Min Winter	2.156	0.00	0.00	100Yr+40% 30Min Summer	90.705	0.00	-0.35
2Yr 1440Min Winter	1.593	0.00	0.00	100Yr+40% 30Min Winter	90.705	0.00	-0.27
2Yr 1440Min Summer	1.593	0.00	0.00	100Yr+40% 60Min Summer	56.713	0.00	-0.17
2Yr 2160Min Winter	1.177	0.00	0.00	100Yr+40% 60Min Winter	56.713	0.00	0.00
2Yr 2160Min Summer	1.177	0.00	0.00	100Yr+40% 120Min Summer	34.246	0.00	0.00
2Yr 2880Min Summer	0.949	0.00	0.00	100Yr+40% 120Min Winter	34.246	0.00	0.00
2Yr 2880Min Winter	0.949	0.00	0.00	100Yr+40% 180Min Summer	25.149	0.00	0.00
2Yr 4320Min Summer	0.701	0.00	0.00	100Yr+40% 180Min Winter	25.149	0.00	0.00
2Yr 4320Min Winter	0.701	0.00	0.00	100Yr+40% 240Min Summer	20.078	0.00	0.00
2Yr 5760Min Summer	0.566	0.00	0.00	100Yr+40% 240Min Winter	20.078	0.00	0.00
2Yr 5760Min Winter	0.566	0.00	0.00	100Yr+40% 360Min Summer	14.585	0.00	0.00
2Yr 7200Min Summer	0.479	0.00	0.00	100Yr+40% 360Min Winter	14.585	0.00	0.00
2Yr 7200Min Winter	0.479	0.00	0.00	100Yr+40% 480Min Summer	11.622	0.00	0.00
2Yr 8640Min Summer	0.419	0.00	0.00	100Yr+40% 480Min Winter	11.622	0.00	0.00
2Yr 8640Min Winter	0.419	0.00	0.00	100Yr+40% 600Min Summer	9.738	0.00	0.00
2Yr 10080Min Winter	0.373	0.00	0.00	100Yr+40% 600Min Winter	9.738	0.00	0.00
2Yr 10080Min Summer	0.373	0.00	0.00	100Yr+40% 720Min Summer	8.424	0.00	0.00
30Yr 15Min Summer	76.035	0.00	-0.16	100Yr+40% 720Min Winter	8.424	0.00	0.00
30Yr 15Min Winter	76.035	0.00	-0.05	100Yr+40% 960Min Summer	6.697	0.00	0.00
30Yr 30Min Summer	49.499	0.00	0.00	100Yr+40% 960Min Winter	6.697	0.00	0.00
30Yr 30Min Winter	49.499	0.00	0.00	100Yr+40% 1440Min Summer	4.839	0.00	0.00
30Yr 60Min Summer	30.811	0.00	0.00	100Yr+40% 1440Min Winter	4.839	0.00	0.00
30Yr 60Min Winter	30.811	0.00	0.00	100Yr+40% 2160Min Summer	3.490	0.00	0.00
30Yr 120Min Summer	18.615	0.00	0.00	100Yr+40% 2160Min Winter	3.490	0.00	0.00
30Yr 120Min Winter	18.615	0.00	0.00	100Yr+40% 2880Min Summer	2.766	0.00	0.00
30Yr 180Min Summer	13.715	0.00	0.00	100Yr+40% 2880Min Winter	2.766	0.00	0.00
30Yr 180Min Winter	13.715	0.00	0.00	100Yr+40% 4320Min Summer	1.989	0.00	0.00
30Yr 240Min Summer	10.995	0.00	0.00	100Yr+40% 4320Min Winter	1.989	0.00	0.00
30Yr 240Min Winter	10.995	0.00	0.00	100Yr+40% 5760Min Summer	1.573	0.00	0.00
30Yr 360Min Summer	8.034	0.00	0.00	100Yr+40% 5760Min Winter	1.573	0.00	0.00
30Yr 360Min Winter	8.034	0.00	0.00	100Yr+40% 7200Min Summer	1.311	0.00	0.00
30Yr 480Min Summer	6.428	0.00	0.00	100Yr+40% 7200Min Winter	1.311	0.00	0.00
30Yr 480Min Winter	6.428	0.00	0.00	100Yr+40% 8640Min Summer	1.129	0.00	0.00
30Yr 600Min Summer	5.404	0.00	0.00	100Yr+40% 8640Min Winter	1.129	0.00	0.00

Simulated Rainfall Events

Storm	Average Intensity (mm/hr)	Runoff Continuity %	Flow Continuity %	Storm	Average Intensity (mm/hr)	Runoff Continuity %	Flow Continuity %
30Yr 600Min Winter	5.404	0.00	0.00	100Yr+40% 10080Min Winter	0.994	0.00	0.00
30Yr 720Min Summer	4.687	0.00	0.00	100Yr+40% 10080Min Summer	0.994	0.00	0.00

Simulation Results

Return Period Yrs: 2.0
Climate Change %: 0

Manholes

Manhole	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
PS12	180 min Summer	115	97.284	0.034	2.056		OK
SA12	15 min Summer	8	97.213	0.108	30.834		OK
PS10	-	0	97.350	0.000	0.000		OK
PS11	-	0	97.350	0.000	0.000		OK
SA22	15 min Summer	8	97.203	0.084	20.759		OK
SA13	15 min Summer	8	96.924	0.127	57.059		OK
SA14	15 min Summer	9	96.173	0.158	73.513		OK
SA15	15 min Summer	9	95.505	0.175	79.123		OK
SA16	15 min Summer	9	95.405	0.175	88.268		OK
PS14	-	0	96.250	0.000	0.000		OK
PS15	-	0	95.750	0.000	0.000		OK
SA17	15 min Summer	9	95.220	0.180	103.349		OK
SA18	15 min Summer	9	94.937	0.230	103.645		OK
SA19	15 min Summer	9	94.778	0.230	103.694		Outfall

Conduits

Pipe No.	Critical Storm	Peak (mins)	US Manhole	DS Manhole	Flow Depth (m)	Max Velocity (m/s)	Max Flow (l/s)	Flow / Capacity	Status
1.000	180 min Summer	115	PS12	SA12	0.034	0.674	2.055	0.116	OK
1.001	15 min Summer	8	SA12	SA13	0.108	1.622	30.594	0.473	OK
2.000	15 min Winter	0	PS10	SA22	0.000	0.000	0.000	0.000	OK
3.000	15 min Winter	0	PS11	SA22	0.000	0.000	0.000	0.000	OK
2.001	15 min Summer	8	SA22	SA13	0.084	1.517	20.607	0.302	OK
1.002	15 min Summer	8	SA13	SA14	0.142	1.719	56.661	0.382	OK
1.003	15 min Summer	9	SA14	SA15	0.167	1.844	74.335	0.560	OK
1.004	15 min Summer	9	SA15	SA16	0.175	1.859	79.561	0.661	OK
1.005	15 min Summer	9	SA16	SA17	0.178	2.032	88.555	0.657	OK
4.000	15 min Winter	0	PS14	SA17	0.000	0.000	0.000	0.000	OK
5.000	15 min Winter	0	PS15	SA17	0.000	0.000	0.000	0.000	OK
1.006	15 min Summer	9	SA17	SA18	0.205	2.010	103.645	0.687	OK
1.007	15 min Summer	9	SA18	SA19	0.230	1.781	103.694	0.961	OK

Permeable Paving Storage

Permeable Paving	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
Permeable Paving4	180 min Summer	115	98.487	0.058	2.068		OK

Return Period Yrs: 30.0

Climate Change %: 0

Manholes

Manhole	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
PS12	120 min Summer	80	97.304	0.054	4.933		OK
SA12	15 min Summer	8	97.263	0.158	55.437		OK
PS10	-	0	97.350	0.000	0.000		OK
PS11	-	0	97.350	0.000	0.000		OK
SA22	15 min Summer	8	97.237	0.117	37.300		OK
SA13	15 min Summer	8	96.977	0.180	102.473		OK
SA14	15 min Summer	10	96.363	0.348	114.114		Surcharged
SA15	15 min Summer	10	95.842	0.512	126.184		Surcharged
SA16	15 min Summer	10	95.736	0.506	139.836		Surcharged
PS14	-	0	96.250	0.000	0.000		OK
PS15	-	0	95.750	0.000	0.000		OK
SA17	15 min Summer	10	95.540	0.500	162.429		Surcharged
SA18	15 min Summer	10	95.169	0.463	163.117		Surcharged
SA19	15 min Summer	10	94.834	0.286	163.344		Outfall

Conduits

Pipe No.	Critical Storm	Peak (mins)	US Manhole	DS Manhole	Flow Depth (m)	Max Velocity (m/s)	Max Flow (l/s)	Flow / Capacity	Status
1.000	120 min Summer	80	PS12	SA12	0.054	0.864	4.932	0.280	OK
1.001	15 min Summer	8	SA12	SA13	0.157	1.849	54.891	0.849	OK
2.000	15 min Summer	0	PS10	SA22	0.000	0.000	0.000	0.000	OK
3.000	15 min Summer	0	PS11	SA22	0.000	0.000	0.000	0.000	OK
2.001	15 min Summer	8	SA22	SA13	0.117	1.769	37.054	0.544	OK
1.002	15 min Summer	9	SA13	SA14	0.239	1.928	101.758	0.686	OK
1.003	15 min Summer	10	SA14	SA15	0.300	2.009	126.465	0.952	Surcharged
1.004	15 min Winter	10	SA15	SA16	0.300	1.909	122.099	1.014	Surcharged
1.005	15 min Winter	10	SA16	SA17	0.300	2.010	135.195	1.003	Surcharged
4.000	15 min Summer	8	PS14	SA17	0.050	0.000	0.000	0.000	OK
5.000	15 min Summer	8	PS15	SA17	0.050	0.000	0.000	0.000	OK
1.006	15 min Winter	10	SA17	SA18	0.300	2.214	156.506	1.038	Surcharged
1.007	15 min Summer	10	SA18	SA19	0.293	2.325	163.344	1.514	OK

Permeable Paving Storage

Permeable Paving	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
Permeable Paving4	120 min Summer	80	98.535	0.106	4.925		OK

Return Period Yrs: 100.0
Climate Change %: 40

Manholes

Manhole	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
PS12	15 min Summer	10	98.314	1.064	1.599		Surcharged
SA12	15 min Summer	10	98.311	1.206	81.391		Flood Risk
PS10	15 min Summer	11	98.127	0.777	0.000		Surcharged
PS11	15 min Summer	11	98.125	0.775	0.000		Surcharged
SA22	15 min Summer	11	98.115	0.996	36.672		Surcharged
SA13	15 min Summer	11	98.031	1.234	113.322		Flood Risk
SA14	15 min Summer	11	97.538	1.523	150.631		Flood Risk
SA15	15 min Summer	11	96.632	1.302	164.714		Flood Risk
SA16	15 min Summer	11	96.445	1.215	183.056		Surcharged
PS14	-	0	96.250	0.000	0.000		OK
PS15	15 min Summer	10	96.109	0.359	1.078		Surcharged
SA17	15 min Summer	10	96.109	1.069	221.354		Surcharged
SA18	15 min Summer	11	95.453	0.747	215.728		Surcharged
SA19	15 min Summer	7	94.848	0.300	168.830		Outfall

Conduits

Pipe No.	Critical Storm	Peak (mins)	US Manhole	DS Manhole	Flow Depth (m)	Max Velocity (m/s)	Max Flow (l/s)	Flow / Capacity	Status
1.000	30 min Summer	21	PS12	SA12	0.150	0.958	8.464	0.480	Surcharged
1.001	15 min Winter	9	SA12	SA13	0.225	1.938	77.039	1.191	OK
2.000	15 min Winter	13	PS10	SA22	0.100	0.670	4.663	0.643	Surcharged
3.000	15 min Winter	13	PS11	SA22	0.100	0.613	4.718	0.609	Surcharged
2.001	15 min Winter	8	SA22	SA13	0.225	1.851	55.566	0.815	OK
1.002	30 min Summer	15	SA13	SA14	0.300	1.987	126.517	0.853	Surcharged
1.003	15 min Winter	11	SA14	SA15	0.300	2.149	151.914	1.144	Surcharged
1.004	15 min Winter	12	SA15	SA16	0.300	2.321	164.084	1.363	Surcharged
1.005	15 min Winter	11	SA16	SA17	0.300	2.543	179.724	1.333	Surcharged
4.000	15 min Summer	6	PS14	SA17	0.050	0.000	0.000	0.000	OK
5.000	15 min Winter	14	PS15	SA17	0.100	0.439	2.192	0.116	Surcharged
1.006	15 min Winter	10	SA17	SA18	0.300	2.971	209.992	1.392	Surcharged
1.007	15 min Winter	11	SA18	SA19	0.300	2.977	210.437	1.951	Surcharged

Permeable Paving Storage

Permeable Paving	Critical Storm	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Flood (m3)	Status
Permeable Paving4	120 min Summer	86	98.622	0.193	7.820		OK